Han Peng

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EDUCATION

2014 – 2018 **Department of Physics, University of Oxford** DPhil (PhD) Reading in Condensed Matter Physics
2010 – 2014 **University of Science and Technology of China (USTC)** Bachelor of Science in Applied Physics

RESEARCH

2018 – Present **Postdoctoral project: Deep Medicine** **Advisor: Prof Steve Smith and Prof Andrei Vedaldi**
Oxford Centre for Functional MRI and Brain, Nuffield Department of Clinical Neuroscience
AND *Visual Geometry Group, Department of Engineering Sciences, University of Oxford*

- **Deep learning in UK Biobank MRI data**
  Looking for disease and aging patterns in ~20000 subjects with 3D MRI data sets

2014 – 2018 **PhD Project: Study of Electronic Properties of Quantum Materials** **Advisor: Prof Yulin Chen**
Clarendon Laboratory, Department of Physics, University of Oxford

- **Machine learning in ARPES data analysis**
  Used convolutional neural network to identify the dispersive features in the spectrum images. The project is developed using MATLAB, Python and TensorFlow
- **Twisted bilayer graphene**
  Used Angular Resolved Photo-Emission Spectroscopy (ARPES) with Micrometre resolution to study electronic properties of twisted bilayer graphene
- **Other contributions**
  Led the development of data analysis system for spectrum image data processing.
  There are 30 users, 4 regular contributors and 10 interns throughout the years. The work includes algorithm optimisation, data visualisation and graphic user interface building

2017 **Research Intern: Neuron Image Registration** **Advisor: Prof Kenneth Harris**
Cortical Processing Laboratory, University College London

- Designed algorithm to register ex-vivo 2D-slices with in-vivo 3D-volume data of neuronal images with different modality and detection methods
- Developed semi-automatic MATLAB data analysis pipeline for the registration algorithm. The pipeline includes data I/O, cell detection, registration, result visualisation and manual modification interface. Project GitHub page: https://git.io/vAjk4

2013 **Research Intern: Study of Jamming Transition in Soft Materials** **Advisor: Prof Ning Xu**
Computational and Theoretical Soft Condensed Matter Physics Laboratory, USTC

- Computationally studied the power law phenomenon in the Jamming transition of a particle system
- Analysed the performance of energy minimization algorithm in different energy potential field

TEACHING

2015 – 2018 Demonstrator (Senior Demonstrator since 2017) in **Computational Physics**, University of Oxford
2017 – 2018 Tutor in **Mechanics**, Jesus College, University of Oxford
2016 – 2017 Tutor in **Condensed Matter Physics**, Hertford College, University of Oxford

AWARDS AND FUNDING

2014 – 2018 Oxford University – China Scholarship Council Scholarship, full scholarship for PhD study
2015 Arthur H. Cooke Memorial Prize awarded by Department of Physics, University of Oxford

TALKS AND PRESENTATIONS

2018.03 **Van hove singularity evolution with twist angle in twisted bilayer graphene**
APS March meeting 2018, Los Angeles, CA, US
2017.11 **Register 2D neuron image with 3D data**
Neuromics project annual retreat 2017, Washington D.C, US

PUBLICATIONS

Total citation: 1472 | h-index: 7 (Nov 2018) | Google Scholar page: https://goo.gl/vlNvHL

Data analysis

1. **H. Peng**, X. Gao, Y. He, Y. W. Li, Y. C. Ji, C. H. Liu, S. A. Ekahana, D. Pei, Z. K. Liu, Z. X. Shen, Y. L. Chen "Super Resolution Convolutional Neural Network for Feature Extraction in Spectroscopic Data" ***Manuscript ready***
2. C. Chen, M. X. Wang,J. X. Wu, Y. Sun, H. F. Yang, Z. Tian, T. Tu, **H. Peng**, G. Li, H. X. Fu, X. Xu, J. Jiang, N. B. M. Schroeter, Y. W. Li, D. Pei, S. Liu, S. Ekahana, H. T. Yuan, J. M. Xue, Z. K. Liu, B. H. Yan, H. L. Peng and Y. L. Chen "Electronic Structures of a High-Mobility Layered Oxychalcogenide Semiconductor, Bi_2O_2Se" ***Science advances, 4 (9), eaat8355 (2018)***
3. S. L. Zhang, W. W. Wang, D. M. Burn, **H. Peng**, H. Berger, A. Bauer, C. Pfleiderer, G. van der Laan, T. Hesjedal "Manipulation of skyrmion motion by magnetic field gradients" ***Nature Communications, 9 (1), 2115 (2018)***

Graphene

4. H. F. Yang, C. Chen, H. Wang, Z. K. Liu, T. Zhang, **H. Peng**, N.B.M. Schröter, S. A. Ekahana, J. Jiang, L. X. Yang, V. Kandyba, A. Barinov, C. Y. Chen, J. Avila, M. C. Asensio, H. L. Peng, Z. F. Liu, and Y. L. Chen "Single Crystalline Electronic Structure and Growth Mechanism of Aligned Square Graphene Sheets" ***APL Materials, 6 (3), 036107 (2018)***
5. **H. Peng**†, N. B. M. Schröter† (equal contribution), J. B. Yin, H. Wang, T.-F. Chung, H. F. Yang, S. Ekahana, Z. K. Liu, J. Jiang, L. X. Yang, T. Zhang, C. Chen, H. Ni, H. Barinov, Y. P. Chen, Z. F. Liu, H. L. Peng, Y. L. Chen "Substrate Doping Effect and Unusually Large Angle van Hove Singularity Evolution in Twisted Bi- and Multilayer Graphene" ***Advanced Materials (2017)***
6. J. B. Yin†, H. Wang†, **H. Peng**† **(equal contribution)**, Z. J. Tan, L. Liao, L. Lin, X. Sun, A. L. Koh, Y. L. Chen**,** H. L. Peng, and Z. F. Liu "Selectively enhanced photocurrent generation in twisted bilayer graphene with van Hove singularity" ***Nature Communications, 7, 10699 (2016)***
7. L. Liao, H. Wang, **H. Peng**, J. B. Yin, A. L. Koh, Y. L. Chen, Q. Xie, H. L. Peng, and Z. F. Liu "van Hove Singularity Enhanced Photochemical Reactivity of Twisted Bilayer Graphene" ***Nano Letters, 15, 5585 (2015)***

Other ARPES related topics

8. J. Jiang, N. B. M. Schröter, S.-C. Wu, N. Kumar, C. Shekhar, **H. Peng**, X. Xu, C. Chen, H. F. Yang, C.-C. Hwang, S.-K. Mo, C. Felser, B. H. Yan, Z. K. Liu, L. X. Yang, and Y. L. Chen "Observation of topological surface states and strong electron/hole imbalance in extreme magnetoresistance compound LaBi" ***Physical Review Materials, 2, 024201 (2018)***
9. A. J. Liang, J. Jiang, M. X. Wang, Y. Sun, N. Kumar, C. Shekhar, C. Chen, **H. Peng**, C. W. Wang, X. Xu, H. F. Yang, S. T. Cui, G. H. Hong, Y. Y. Y. Xia, S. K. Mo, Q. Gao, X. J. Zhou, L. X. Yang, C. Felser, B. H. Yan, Z. K. Liu, Y. L. Chen 2017. Observation of the topological surface state in the nonsymmorphic topological insulator KHgSb. ***Physical Review B, 96(16), p.165143 (2017)***
10. J. Jiang, Z. K. Liu, Y. Sun, H. F. Yang, R. Rajamathi, Y. P. Qi, L. X. Yang, C. Chen, **H. Peng**, C.-C. Hwang, S. Z. Sun, S.-K. Mo, I. Vobornik, J. Fujii, S. S. P. Parkin, C. Felser, B. H. Yan, Y. L. Chen "Signature of type-II Weyl semimetal phase in MoTe2" ***Nature Communications, 8, 13973 (2017)***
11. Z. K. Liu, L. X. Yang, Y. Sun, T. Zhang, **H. Peng**, H. F. Yang, C. Chen, Y. Zhang, Y. F. Guo, D. Prabhakaran, M. Schmidt, Z. Hussain, S.-K. Mo, C. Felser, B. Yan and Y. L. Chen "Evolution of the Fermi surface of Weyl semimetals in the transition metal pnictide family" ***Nature Materials, 15, 27 (2016)***
12. L. X. Yang, Z. K. Liu, Y. Sun, **H. Peng**, H. F. Yang, T. Zhang, B. Zhou, Y. Zhang, Y. F. Guo, M. Rahn, D Prabhakaran, Z. Hussain, S.–K. Mo, C. Felser, B. Yan and Y. L. Chen "Weyl Semimetal Phase in non-Centrosymmetric Compound TaAs" ***Nature Physics, 11, 728 (2015)***
13. Z. K. Liu, J. Jiang, B. Zhou, Z. J. Wang, Y. Zhang, H. M. Weng, D. Prabhakaran, S. -K. Mo, **H. Peng**, P. Dudin, T. Kim, M. Hoesch, Z. Fang, X. Dai, Z. X. Shen, D. L. Feng, Z. Hussain, Y. L. Chen. "A Stable Three-dimensional Topological Dirac Semimetal Cd3As2" ***Nature Materials, 13, 677 (2014)***

REFERENCE

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Prof Kenneth Harris *Institute of Neurology, University College London* Kenneth.Harris@ucl.ac.uk
Prof Thorsten Hesjedal *Physics Department, University of Oxford* Thorsten.Hesjedal@physics.ox.ac.uk